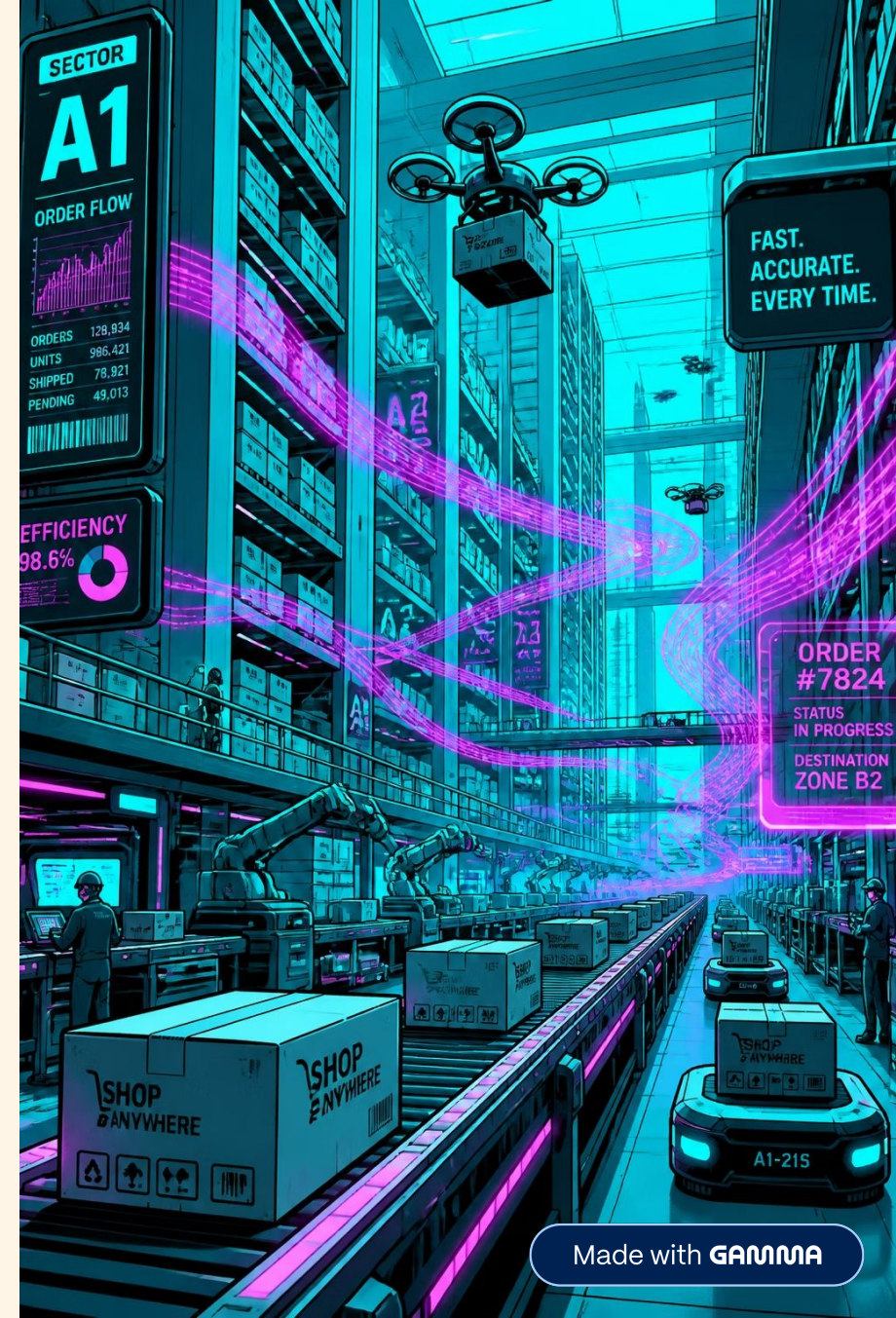


Intelligent E-Commerce Product Sales Forecasting

A Machine Learning Approach to Demand Prediction

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Institution: Lovely Professional University



Made with GAMMA

Project Objective & Business Problem

The Problem

E-commerce platforms struggle with overstocking and understocking due to unpredictable consumer demand.

The Solution

Developing an automated, data-driven machine learning model to accurately forecast future product sales.

The Goal

Optimize supply chain operations, reduce inventory costs, and ensure high-demand products remain in stock.



The Technology Stack



Core Language

Python.



Data Processing

Data manipulation utilizing pandas and numpy.



Machine Learning

Predictive modeling driven by the catboost algorithm.



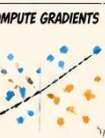
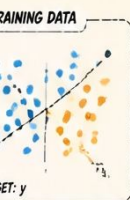
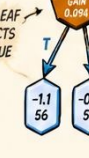
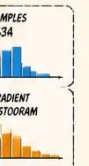
Visualization

Advanced graphing and interactive plotting using matplotlib, plotly, and graphviz.

BUILD STRONG PREDICTIVE MODELS ONE TREE AT A TIME



- initialize model with constant prediction
- compute gradients of the loss function
- fit a decision tree to the negative gradients
- update predictions
- repeat for M trees



2. FIT TREE TO GRADIENTS

3. UPDATE PREDICTIONS

4. REDUCE RESIDUALS

5. REPEAT.

1 Highly Efficient Gradient Boosting

CatBoost (Categorical Boosting) is a highly efficient, open-source gradient boosting algorithm.

2 Native Categorical Handling

It natively handles categorical variables without requiring extensive data preprocessing like one-hot encoding.

3 Symmetric Trees

It utilizes symmetric trees to prevent overfitting and improve prediction speed.

Model Training & Experimentation

01

Iterative Experiments

Multiple iterative experiments were conducted, tracking different feature sets, including the baseline ProductSalesModel and iterations like `new_features_1`.

02

Training Cycles

The models were configured to train over extensive cycles, specifically utilizing 500 to 1000 total iterations.

03

Continuous Monitoring

Training involved continuous monitoring of learning rates and elapsed time to balance accuracy with computational efficiency.

Evaluation Metric (RMSE)

What is RMSE?

The primary evaluation metric utilized for optimizing the model was Root Mean Squared Error (RMSE).

RMSE was strictly monitored across both the learning sets and the testing sets.

Results

Across the experiments, the model demonstrated strong convergence, achieving test RMSE scores in the **~35.7 to ~38.6 range**, indicating reliable prediction accuracy.

Explainable AI (XAI) Integration

The Challenge

Machine learning models are often "black boxes," but this project prioritizes transparency.

The Pipeline

The pipeline integrates an AdditiveForceVisualizer to break down how individual features influence the final prediction.

Stakeholder Visibility

This allows stakeholders to see exactly which variables are driving sales up or down for any given product.



Interactive Feature Visualization

Feature Effects Mapping

The visualizer maps feature effects against a centralized base value.

Contrasting Color Maps

It utilizes contrasting color maps (such as Red/Blue distributions) to clearly delineate positive and negative impacts on the sales forecast.

Dynamic Scaling

The custom interface scales dynamically, providing precise numeric formatting for the effect of each feature.



Business Value & Application



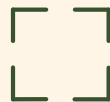
Proactive Planning

Allows warehouse managers to adjust stock levels weeks in advance based on the AI's numerical outputs.



Marketing Alignment

Helps teams understand which specific features (identified by the force visualizer) are correlating with sales spikes.



Scalability

The pipeline can be scaled across thousands of unique stock codes.

Conclusion & Q&A

1

Accurate & Scalable Framework

CatBoost provides a highly accurate and scalable framework for e-commerce forecasting.

2

Bridging the Gap

Integrating force visualization bridges the gap between complex mathematical outputs and actionable business insights.



Questions?